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Productivity Losses from Financial Frictions: Can Self-Financing Undo Capital Misallocation?

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1 Research Question, Contribution, and Roadmap

1.1 What is being asked?

The paper asks a very specific question:

If entrepreneurs face borrowing constraints, can they *self-finance* their desired capital over time by saving out of profits, so that capital misallocation and measured TFP losses vanish? If yes, how fast does this happen, and what features of the productivity process govern the speed?

1.2 Core contribution

The paper shows that the evolution of *aggregate* TFP can be written as a simple functional of *wealth shares by productivity type*, and those wealth shares satisfy a tractable PDE; the extent and speed of “self-financing undoing misallocation” are governed by the persistence of idiosyncratic productivity shocks.

1.3 How tractability is achieved

Two modeling choices generate closed-form aggregation:

1. **Constant returns to scale (CRS) technology** plus competitive factor markets $\Rightarrow$ profits are *linear* in capital and hence (under the collateral constraint) linear in wealth.
2. **Log utility in continuous time** $\Rightarrow$ an *affine* consumption rule $c = \rho a$ and therefore *linear* wealth dynamics $\dot{a} = s(z, t)a$.

Linearity makes it possible to replace the full joint distribution $g_t(a, z)$ with the lower-dimensional object $\omega(z, t)$: *wealth shares across productivity types*.

2 Environment

Time is continuous, $t \in [0, \infty)$. There is a continuum of entrepreneurs indexed by idiosyncratic productivity z and wealth a .

2.1 Preferences

Each entrepreneur has exponential discounting and log utility:

$$\max \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} \log c_t \, dt \right], \quad \rho > 0. \quad (1)$$

2.2 Technology

An entrepreneur with productivity z chooses capital k and labor ℓ to produce

$$y = f(z, k, \ell) = (zk)^\alpha \ell^{1-\alpha}, \quad \alpha \in (0, 1), \quad (2)$$

with depreciation rate $\delta > 0$.

2.3 Markets and prices

There are competitive markets for:

- labor, with wage $w(t)$,
- rented capital, with net interest rate $r(t)$ and rental rate $R(t) = r(t) + \delta$.

Workers supply an exogenous amount L of labor and are assumed to be hand-to-mouth (no saving). This pins down the closure: aggregate assets/capital are owned (and accumulated) by entrepreneurs.

2.4 Wealth dynamics and collateral constraint

Let $a(t)$ denote an entrepreneur's financial wealth (net worth). The flow budget constraint is

$$\dot{a} = f(z, k, \ell) - w\ell - (r + \delta)k + ra - c. \quad (3)$$

Financial frictions take the reduced-form collateral form

$$k \leq \lambda a, \quad \lambda \geq 1. \quad (4)$$

3 Individual Optimization

3.1 Static factor demands and profits

Given prices (w, r) , define the constrained profit function

$$\Pi(a, z) \equiv \max_{k, \ell} \{f(z, k, \ell) - w\ell - (r + \delta)k \text{ s.t. } k \leq \lambda a\}. \quad (5)$$

Lemma 1 (Factor demands and linear profits). *Define*

$$\pi \equiv \alpha \left(\frac{1 - \alpha}{w} \right)^{\frac{1-\alpha}{\alpha}}, \quad \underline{z} \text{ solves } \underline{z}\pi = r + \delta.$$

Then the entrepreneur is either inactive or fully levered:

$$k(a, z) = \begin{cases} \lambda a, & z \geq \underline{z}, \\ 0, & z < \underline{z}, \end{cases} \quad (\text{L1-k})$$

$$\ell(a, z) = \left(\frac{1-\alpha}{w} \right)^{\frac{1}{\alpha}} z k(a, z), \quad (\text{L1-l})$$

$$\Pi(a, z) = \max\{z\pi - r - \delta, 0\} \lambda a. \quad (\text{L1-Pi})$$

Hence profits are linear in wealth a .

Proof (Appendix A1 logic). Fix (a, z) and first choose ℓ conditional on (k, z) :

$$\max_{\ell \geq 0} (zk)^\alpha \ell^{1-\alpha} - w\ell \quad \Rightarrow \quad (1-\alpha)(zk)^\alpha \ell^{-\alpha} = w,$$

so

$$\ell^*(z, k) = \left(\frac{1-\alpha}{w} \right)^{1/\alpha} zk.$$

Substituting back yields the maximized operating profit (before paying for capital)

$$(zk)^\alpha (\ell^*)^{1-\alpha} - w\ell^* = z\pi k.$$

Thus the profit problem becomes

$$\Pi(a, z) = \max_k \{z\pi k - (r + \delta)k\} \quad \text{s.t.} \quad k \leq \lambda a,$$

which is linear in k , hence $k \in \{0, \lambda a\}$. The cutoff $\underline{z}$ solves $z\pi = r + \delta$. □

3.2 Dynamic consumption–saving problem (HJB)

Using Lemma 1, the wealth dynamics can be written as

$$\dot{a} = \Pi(a, z) + ra - c = A(z, t)a - c, \quad A(z, t) \equiv \lambda \max\{z\pi(t) - r(t) - \delta, 0\} + r(t).$$

The entrepreneur's HJB is

$$\rho V(a, z, t) = \max_{c \geq 0} \left\{ \log c + \frac{1}{dt} \mathbb{E}[dV(a, z, t)] \right\} \quad \text{s.t.} \quad da = [A(z, t)a - c] dt.$$

Lemma 2 (Linear saving rate under log utility). *The optimal policy satisfies*

$$c = \rho a, \quad \dot{a} = s(z, t) a, \quad s(z, t) = \lambda \max\{z\pi(t) - r(t) - \delta, 0\} + r(t) - \rho. \quad (6)$$

Proof (Appendix A2 logic). Guess $V(a, z, t) = \rho^{-1}(v(z, t) + \log a)$. Then $V_a = (\rho a)^{-1}$. The FOC for c in the HJB gives $1/c = V_a$, so $c = \rho a$. Substituting into $\dot{a} = Aa - c$ yields $\dot{a} = [A(z, t) - \rho]a$.

□

4 Equilibrium and Aggregation

4.1 Wealth shares

Let $G_t(a, z)$ be the joint distribution of wealth and productivity across entrepreneurs at time t , with density $g_t(a, z)$ when it exists. Define aggregate entrepreneurial wealth (equal to the aggregate capital stock in this economy)

$$K(t) \equiv \int a \, dG_t(a, z).$$

Define the *wealth share density* of productivity type z as

$$\omega(z, t) \equiv \frac{1}{K(t)} \int_0^\infty a g_t(a, z) \, da, \quad \int_0^\infty \omega(z, t) \, dz = 1. \quad (9)$$

Let $\Omega(z, t) \equiv \int_0^z \omega(x, t) \, dx$.

4.2 Market clearing and the cutoff

Capital market clearing is

$$\int k(a, z, t) \, dG_t(a, z) = \int a \, dG_t(a, z) = K(t). \quad (7)$$

Labor market clearing is

$$\int \ell(a, z, t) \, dG_t(a, z) = L. \quad (8)$$

Using Lemma 1, $k(a, z, t) = \lambda a$ for active entrepreneurs ($z \geq \underline{z}(t)$) and 0 otherwise. Thus (7) implies the cutoff condition

$$\lambda(1 - \Omega(\underline{z}(t), t)) = 1. \quad (14)$$

Only the top $1/\lambda$ share of wealth (by z -type) is *active* in equilibrium.

4.3 Aggregation: GDP, TFP, and capital accumulation

Proposition 1 (Aggregate production and capital dynamics). *Given a path of wealth shares $\omega(z, t)$, equilibrium aggregates satisfy*

$$Y(t) = Z(t) K(t)^\alpha L^{1-\alpha}, \quad (11)$$

$$\dot{K}(t) = \alpha Z(t) K(t)^\alpha L^{1-\alpha} - (\rho + \delta)K(t), \quad (12)$$

where measured TFP is

$$Z(t) = \left(\frac{\int_{\underline{z}(t)}^\infty z \omega(z, t) \, dz}{1 - \Omega(\underline{z}(t), t)} \right)^\alpha = (\mathbb{E}_{\omega(\cdot, t)}[z \mid z \geq \underline{z}(t)])^\alpha. \quad (13)$$

Factor prices satisfy

$$w(t) = (1 - \alpha) Z(t) K(t)^\alpha L^{-\alpha}, \quad (15-w)$$

$$r(t) = \alpha \zeta(t) Z(t) K(t)^{\alpha-1} L^{1-\alpha} - \delta, \quad \zeta(t) \equiv \frac{\underline{z}(t)}{\mathbb{E}_{\omega(\cdot, t)}[z \mid z \geq \underline{z}(t)]} \in [0, 1]. \quad (15-r)$$

Interpretation. Equation (13) is the paper’s central “TFP decomposition”: TFP is a truncated wealth-weighted mean of productivity among *active* entrepreneurs. Financial development λ matters because it affects the cutoff $\underline{z}(t)$ and the endogenous object $\omega(z, t)$.

5 Dynamics of Wealth Shares

5.1 Idiosyncratic productivity diffusion

Productivity follows an ergodic diffusion

$$dz = \mu(z) dt + \sigma(z) dW_t. \quad (20)$$

5.2 A PDE for wealth shares

Proposition 2 (Law of motion for wealth shares). *Wealth shares satisfy*

$$\frac{\partial \omega(z, t)}{\partial t} = \left[s(z, t) - \frac{\dot{K}(t)}{K(t)} \right] \omega(z, t) - \frac{\partial}{\partial z} [\mu(z) \omega(z, t)] + \frac{1}{2} \frac{\partial^2}{\partial z^2} [\sigma^2(z) \omega(z, t)]. \quad (21)$$

In a stationary equilibrium, ω solves

$$0 = s(z) \omega(z) - \frac{d}{dz} [\mu(z) \omega(z)] + \frac{1}{2} \frac{d^2}{dz^2} [\sigma^2(z) \omega(z)]. \quad (22)$$

Interpretation. Equation (21) has three parts:

1. **Differential accumulation:** if $s(z, t) > \dot{K}/K$, the wealth share of type z rises.
2. **Type mobility (drift):** $\mu(z)$ moves mass across z .
3. **Type mobility (diffusion):** $\sigma(z)$ spreads mass across z .

The first term is a “replicator” term: wealth shares grow where individual wealth grows faster than aggregate wealth.

6 Steady State and Misallocation

6.1 Steady state definition and aggregates

A steady state is

$$\dot{K}(t) = 0, \quad \omega(z, t) = \omega(z), \quad r(t) = r, \quad w(t) = w. \quad (16)$$

From Proposition 1, steady-state aggregates satisfy

$$Y = ZK^\alpha L^{1-\alpha}, \quad (17)$$

$$\alpha ZK^{\alpha-1} L^{1-\alpha} = \rho + \delta, \quad (18)$$

implying

$$\frac{K}{Y} = \frac{\alpha}{\rho + \delta}. \quad (19)$$

6.2 A wedge representation

The wage equals the aggregate MPL, but the rental rate equals the MPK of the *marginal* entrepreneur:

$$r + \delta = \underline{z} \pi.$$

Define the misallocation wedge

$$\zeta \equiv \frac{\underline{z}}{\mathbb{E}_\omega[z \mid z \geq \underline{z}]} \in [0, 1].$$

If capital were efficiently allocated among active entrepreneurs, $\underline{z}$ would equal the active average and $\zeta = 1$.

7 Persistence as the Key Primitive

7.1 Varying persistence without changing the stationary cross-section

To change *persistence* while holding the stationary cross-sectional distribution fixed, the paper scales the diffusion by $\theta > 0$:

$$dz = \frac{1}{\theta} \tilde{\mu}(z) dt + \sqrt{\frac{1}{\theta}} \tilde{\sigma}(z) dW_t. \quad (25)$$

A canonical example is an OU process for $\log z$:

$$d \log z = -\frac{1}{\theta} \log z dt + \sigma \sqrt{\frac{1}{\theta}} dW_t, \quad (26)$$

implying $\text{corr}(\log z_t, \log z_{t+s}) = e^{-s/\theta}$.

7.2 How θ enters the stationary wealth share equation

Under (25), the stationary equation (22) becomes

$$0 = \theta s(z) \omega(z) - \frac{d}{dz} [\tilde{\mu}(z) \omega(z)] + \frac{1}{2} \frac{d^2}{dz^2} [\tilde{\sigma}^2(z) \omega(z)]. \quad (28)$$

Thus larger θ makes differential wealth accumulation more important relative to type mobility.

7.3 Main comparative static (statement)

Proposition 3 (TFP is increasing in persistence (statement)). *Fix $\lambda \in [1, \infty)$. Under mild regularity conditions, steady-state measured TFP $Z(\theta)$ is continuous and strictly increasing in θ . Moreover,*

$$Z(0) = (\mathbb{E}_\psi[z \mid z \geq \underline{z}(0)])^\alpha, \quad \lim_{\theta \rightarrow \infty} Z(\theta) = \bar{z}^\alpha,$$

where ψ is the stationary distribution of z and $\bar{z}$ is the upper bound of the support.

Intuition. Self-financing needs sustained time in a high- z state. If z is transitory, high- z entrepreneurs do not remain productive long enough to accumulate wealth and relax their constraints. If z is persistent, productive entrepreneurs save and eventually dominate the wealth distribution, raising Z toward the first best.

8 Transition Experiments (What is simulated)

The transition analysis in the paper solves the PDE (21) forward from initial conditions. Two key ingredients are (i) a starting wealth-share distribution $\omega_0(z)$ and (ii) a productivity process (persistence).

8.1 Parameterized initial wealth shares

For two of the three transition experiments, initial wealth shares are chosen as a shifted log-normal:

$$\omega_0(z) \propto \frac{1}{z} \exp\left(-\frac{(\log z - m)^2}{\sigma^2}\right). \quad (29)$$

- $m = 0$ implies $\omega_0(z)$ coincides with the stationary productivity distribution $\psi(z)$: wealth is initially *independent* of productivity.
- $m < 0$ implies initial wealth is tilted toward low productivity (“distorted”).
- $m > 0$ implies initial wealth is tilted toward high productivity (“less distorted”).

8.2 Three experiments

1. **Buera–Shin style reform:** a reform removes idiosyncratic distortions that are correlated with z .
2. **Distorted initial allocation:** $m = -0.5$ (wealth negatively correlated with productivity).
3. **Less distorted initial allocation:** $m = 0.25$ (wealth positively correlated with productivity).

8.3 Speed metric: half-life of $Z(t)$

The paper summarizes speed using the half-life of the TFP transition: the time it takes for $Z(t)$ to close half of the gap between its initial and steady-state levels. Figure 3 shows that this half-life increases sharply with persistence.

9 Figures: Interpretation (Figures 1–6)

9.1 Important note about “tables”

The provided AER article PDF contains **no numbered tables**. The text references *Table E1* (external finance-to-GDP for India), but that table is in the paper’s *online appendix* and is not included in the attached PDF. Accordingly, these notes interpret **all figures in the PDF (Figures 1–6)** and all quantitative claims reported in the main text.

9.2 Figure 1: Wealth shares vs. productivity distribution (steady state)

555**What is plotted.** For different values of the autocorrelation of $\log z$, the figure plots:

- the stationary productivity density $\psi(z)$ (dashed), and
- the stationary wealth share density $\omega(z)$ (solid).

Interpretation. Low persistence implies $\omega \simeq \psi$ (little selection). High persistence implies ω shifts to high z (strong selection/self-financing).

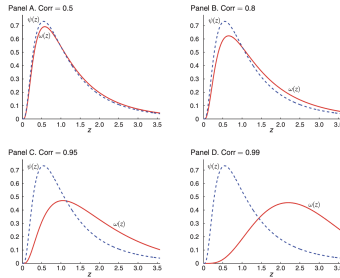


FIGURE 1. WEALTH SHARES AND AUTOCORRELATION
Notes: The dashed lines are the productivity distribution $\psi(z)$ from (27). The solid lines are the wealth shares $\omega(z)$; i.e., the solution to (22) for the stochastic process (26). As persistence θ (equivalently autocorrelation) increases, wealth becomes more concentrated with high-productivity entrepreneurs.

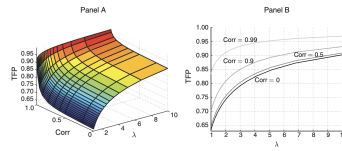


FIGURE 2. TFP AND AUTOCORRELATION
Notes: Panel B displays a cross-section of the three-dimensional graph in panel A. Again, note the sensitivity in the range $\text{corr} = 0.75$ to $\text{corr} = 1$. Parameters are $\alpha = 1/\lambda$, $\rho = 0.05$, $\sigma_V = \log(0.85) = 0.56$, and Γ vary $\text{corr} = \exp(-(1/\theta))$.

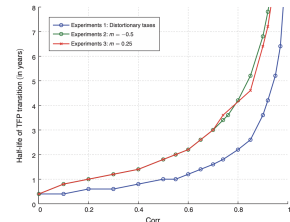


FIGURE 3. EFFECT OF AUTOCORRELATION ON SPEED OF TRANSITION

9.3 Figure 2: TFP depends jointly on persistence and λ

Panel A shows Z as a surface in $(\text{autocorr}, \lambda)$. **Panel B** shows slices: $Z(\lambda)$ for various autocorr values.

Interpretation. Relaxing constraints (higher λ) increases TFP, but the marginal gain is much larger when persistence is low (because self-financing is weak).

9.4 Figure 3: Persistence slows transitions (half-lives)

The half-life of $Z(t)$ is increasing in autocorrelation for all three experiments.

Interpretation. Persistent productivity makes long-run selection stronger (good) but reallocating wealth across types slower (slow transitions).

9.5 Figure 4: Full transition paths for $Z(t)$, $K(t)$, GDP, and $r(t)$

Higher persistence yields higher long-run Z and GDP but slower convergence. In early years, a more persistent economy can look worse despite a better asymptote.

9.6 Figure 5: The transition is the evolution of wealth shares

With i.i.d. shocks, $\omega(z, t)$ jumps quickly to its stationary shape. With persistent shocks, $\omega(z, t)$ gradually shifts to higher z over many years.

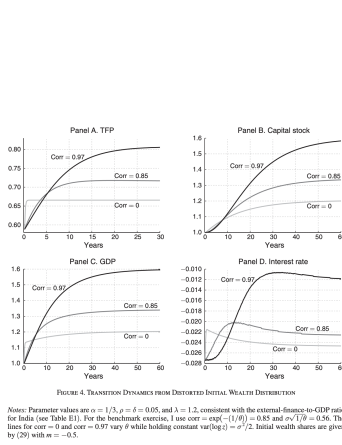


FIGURE 4. TRANSITION DYNAMICS FROM DISTORTED INITIAL WEALTH DISTRIBUTION

Notes: Parameter values are $\alpha = 1/3$, $\beta = 0.05$, and $\lambda = 1.2$, consistent with the external finance-to-GDP ratio for India (see Table E1). For the benchmark exercise, $1/\text{corr} = \exp(-1/\theta) = 0.85$ and $\rho_V/\beta = 0.56$. The lines for $\text{corr} = 0$ and $\text{corr} = 0.97$ vary θ while holding constant $\text{var}(\log z) = \sigma^2/2$. Initial wealth shares are given by (29) with $m = -0.5$.

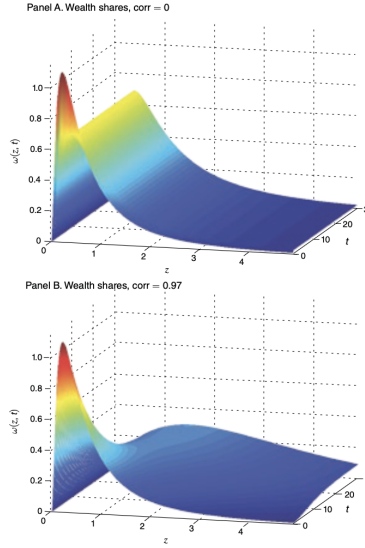


FIGURE 5. TRANSITION DYNAMICS FOR WEALTH SHARES

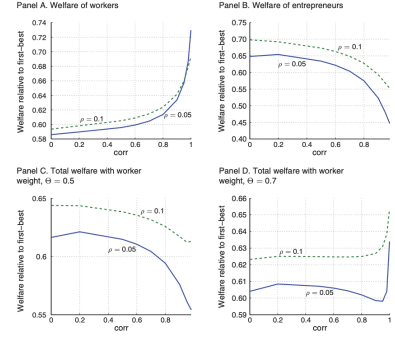


FIGURE 6. WELFARE LOSSES

Notes: Parameter values are $\alpha = 1/3$, $\beta = 0.05$, and $\lambda = 1.2$, consistent with the external finance-to-GDP ratio for India (see Table E1), and $1/\text{corr} = \exp(-1/\theta)$ while holding constant $\text{var}(\log z) = \sigma^2/2 = (0.56)/(-2\log(0.85)) = 0.96$. Initial wealth shares are the same as in Figure 4 and given by (29) with $m = -0.5$. Welfare relative to first-best is in units of permanent consumption: $C_0/(1-\rho_V) - \exp(\rho_V/V_0 - V_0^*)$ where V_0 is defined in (24).

9.7 Figure 6: Welfare depends on whose welfare

Workers benefit from higher long-run output, but discounting makes them care about short-run paths too. Entrepreneurs face an additional trade-off: higher persistence raises average returns but reduces their ability to self-insure (more persistent bad times). Thus entrepreneurial welfare can be non-monotone, and aggregate welfare depends on the worker weight Θ .

10 Welfare Accounting (Lemma 3)

Let $\Theta \in [0, 1]$ be the Pareto weight on workers. Welfare at time t is

$$\bar{V}_t = (1 - \Theta)\bar{V}_t^E + \Theta\bar{V}_t^W. \quad (24)$$

Entrepreneur welfare decomposes as

$$\bar{V}_t^E = \frac{1}{\rho} \left(\int_0^\infty v(z, t) \psi_t(z) \, dz + \int_0^\infty \log(a) \varphi_t(a) \, da \right), \quad (24-E)$$

$$\bar{V}_t^W = \int_t^\infty e^{-\rho(s-t)} \log[w_s L] \, ds, \quad (24-W)$$

where ψ_t and φ_t are the marginals of G_t , and $v(z, t)$ is the z -component of the entrepreneur value function.

11 Sensitivity and Extensions (as discussed in the paper)

11.1 CRRA utility

With CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$, the exact log-utility aggregation breaks, but the paper argues that (for γ near 1) the qualitative persistence logic survives. A key modification is a time-varying aggregate MPC

$$\tilde{\rho}(t; \gamma, \lambda) \equiv \int_0^\infty \hat{c}(z, t; \gamma, \lambda) \omega(z, t) \, dz, \quad (30)$$

which weakly re-introduces a direct role for λ in K/Y .

11.2 Decreasing returns

With decreasing returns to scale at the firm level, profits and factor demands are not linear in wealth, and $Z(t)$ cannot be written as the simple truncated mean (13). The CRS case is treated as a tractable limiting case (as $\nu \rightarrow 1$), suggesting continuity of the main mechanism.

12 Short Takeaways

1. **Misallocation is about who holds wealth.** TFP depends on wealth-weighted productivity among active entrepreneurs.
2. **Wealth shares obey a PDE.** The model reduces to tracking $\omega(z, t)$ rather than the full $g(a, z, t)$.
3. **Persistence governs both levels and speeds.** More persistence $\Rightarrow$ better steady state but slower transitions.
4. **Welfare is nuanced.** Worker welfare tends to rise with persistence; entrepreneurial welfare may be hump-shaped.

13 Conclusion

13.1 Summary

This paper develops a highly tractable general-equilibrium model of heterogeneous entrepreneurs who face **collateral constraints** but can **self-finance** (accumulate wealth) through forward-looking savings. The model is analytically disciplined enough that aggregate output can be represented as a **Solow-style aggregate production function with *time-varying* TFP**, where TFP is pinned down by the evolving joint distribution of entrepreneurial productivity and wealth.

The paper’s central conclusion is that the **persistence of idiosyncratic productivity shocks** is the key state variable that jointly determines:

1. the **magnitude of steady-state productivity losses** from financial frictions (capital misallocation), and
2. the **speed of transition dynamics** (how quickly an economy converges after a reform or a change in financial conditions).

Specifically, the paper establishes a sharp **tradeoff**:

- If productivity shocks are **highly persistent**, then **steady-state TFP losses are small** because self-financing gradually concentrates wealth (and therefore capital) in the hands of high-productivity entrepreneurs, pushing the allocation toward efficiency. However, the **transition is slow**: misallocation unwinds only gradually because wealth accumulation is slow, so TFP evolves endogenously over time.
- If productivity shocks are **transitory** (close to i.i.d.), then **steady-state TFP losses are large** because entrepreneurs cannot remain productive long enough to accumulate sufficient internal funds to finance their desired scale; wealth does not become strongly aligned with productivity. But the **transition is fast**: the economy reaches its (inefficient) steady state quickly, and TFP adjusts almost immediately.

A key policy implication follows immediately: **steady-state comparisons can be misleading**. Even if financial frictions appear “small” in the long run (under persistent shocks), they can still be **first-order in the short run** because the economy may spend a long time in transition, with sustained (endogenous) TFP dynamics.

13.2 Economic intuition: why persistence governs both long-run losses and adjustment speed

The mechanism is easiest to see from the paper’s core aggregation logic:

1. **Collateral constraints tie scale to wealth.** With a borrowing limit, an entrepreneur’s capital choice is effectively bounded by their wealth. This creates misallocation when high-productivity entrepreneurs are poor and low-productivity entrepreneurs are rich.

2. **Self-financing aligns wealth with productivity, but only through time.** Entrepreneurs accumulate wealth out of profits (past successes). Wealth therefore becomes a state variable summarizing past productivity histories.
3. **Persistence determines whether “success” lasts long enough to matter.** If productivity is persistent, a high-productivity entrepreneur stays productive long enough to generate profits and build wealth, so over time wealth shares tilt toward high-productivity types (Figure 1 and Figure 5). This progressively relaxes constraints for the right entrepreneurs and improves the allocation, raising TFP.
4. **But the same force makes adjustment slow.** The improvement requires wealth to move across the cross-section, and wealth moves only through saving. When productivity is persistent, the entire *joint* distribution of productivity and wealth adjusts slowly, so TFP exhibits long, endogenous transition dynamics (Figure 3 and Figure 4).
5. **When productivity is i.i.d., re-shuffling breaks the wealth–productivity link.** If productivity is essentially re-drawn every period, then today’s high-productivity entrepreneurs are not systematically yesterday’s winners. Self-financing cannot reliably target capital toward productive types, so the steady state is inefficient. Yet because the cross-sectional link between wealth and productivity becomes irrelevant quickly, the economy reaches that inefficient steady state rapidly (Figure 5, Panel A).

13.3 Quantitative and graphical takeaways (how to remember the results)

- **Figure 2 (TFP and autocorrelation):** steady-state TFP increases with both credit-market tightness relief (higher λ) and shock persistence (higher corr), and it is **highly sensitive** as $\text{corr} \rightarrow 1$. Intuition: sufficiently persistent productivity allows self-financing to approximate the efficient allocation even with limited external finance.
- **Figure 3 (speed of transition):** the half-life of TFP convergence rises sharply with persistence. Intuition: persistent shocks generate persistent wealth-share dynamics, hence persistent TFP dynamics.
- **Figure 4 (transition paths):** with more persistent shocks, **long-run** TFP is higher (smaller long-run losses), but **short-run** TFP can be lower for longer (larger short-run losses) because the initial misallocation unwinds slowly. This is exactly the scenario in which steady-state analysis understates the relevance of financial frictions.
- **Figure 6 (welfare):** persistence affects welfare through two opposing forces: (i) higher persistence tends to raise productivity and returns (a level effect), but (ii) more persistent idiosyncratic risk is harder to self-insure (a risk/insurance effect). Consequently, entrepreneurial welfare need not be monotone in persistence, and total welfare depends on the planner’s distributional weights.

13.4 What the paper contributes conceptually

1. **A clean separation of long-run vs short-run:** the paper emphasizes that “finance and misallocation” cannot be assessed by steady states alone; the transition path is endogenous and can be long-lived.
2. **A reconciliation channel across literatures:** differences in assumed shock persistence help explain why otherwise similar quantitative models of financial frictions can generate very different TFP-loss estimates.
3. **A tractable mapping from micro frictions to macro TFP:** aggregate TFP is tightly linked to wealth shares across productivity types; solving the equilibrium reduces to solving for the evolution of those shares, providing tools (e.g., the paper’s propositions on wealth-share dynamics) that can be portable to other applications.

13.5 Takeaway

In a collateral-constrained economy with self-financing, the persistence of idiosyncratic productivity shocks governs everything: with persistent shocks, self-financing eventually aligns wealth with productivity so long-run TFP losses are small, but convergence is slow and short-run losses can be large; with transitory shocks, long-run losses are large but the economy reaches them quickly.